



Shri Guru Gobind Singh Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Production Engineering
End Semester Examination (ESE)
Academic Year: 2025-26 (Semester I)

Class: TY B.Tech.

Course: Smart Materials

Course Code: PEC-PE305

Date: 26.11.2025

Time: 10:00 to 12:00

Max. Marks: 60

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.		Questions	Marks	CO	BT
Q.1	A	Write the importance and future possibilities of smart materials in your words.	5	CO2	BT2
	B	What is meant by self-healing materials? How it works?	5	CO4	BT2
Q.2	A	Explain in detail the Magneto strictive Materials along with properties, governing relations, strength and weaknesses.	5	CO4	BT3
	B	Illustrate how Electroactive Polymers works, along with their properties, governing relations, strength and weaknesses.	5	CO4	BT3
Q.3	A	Write detailed note on Ferrofluids and Magneto rheological Fluids.	5	CO3	BT2
	B	What is Soft Matter and its applications as smart skins, smart textiles, write in details.	5	CO4	BT2
Q.4	A	What are different energy storage systems?	3	CO5	BT1
	B	What is Hydrogen Storage Materials? What are the future challenges in this technology?	7	CO5	BT4
Q.5		Explain in detail about manufacturing technique of any ONE: LIGA process OR STL process.	10	CO3	BT4
Q.6		What is generative manufacturing techniques? Explain any TWO of the following: SLS, BPM, LOM, FDM, BIS	10	CO4	BT4

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Classify and select different types of smart materials.
CO 2	Comprehend Important Concepts and principles of Smart Materials.
CO 3	Synthesis, sensing and actuation of Piezoelectric Materials, Magneto strictive Materials, Shape Memory Alloys, Electroactive Polymers
CO 4	Synthesis, sensing and actuation of Ferrofluids and Magneto rheological Fluids, Soft Matter, Carbon Nanotubes and Carbon nanostructures, Thermoelectric Materials
CO 5	Classify and select Smart Materials for Energy Applications: Materials used for energy storage.



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Production Engineering.
End Semester Examination
Academic Year: 2025-26

Class: TY B.Tech.

Date: 21.11.2025

Course: Operations Research.

Time: 10:00 to 12:00

Course Code: PCC-PE305

Max. Marks: 60

Note:

- Assume suitable data whenever necessary and state it clearly.
- Solve **any SIX questions** from questions 1 to 7.
- Figures to right indicate full marks.
- Paper containing 2 pages.

Q.1		Answers the following. (Any 2)																								
	a	What is OR? Describe briefly its application.	05	CO1	BT1																					
	b	Explain methodology of OR.	05	CO1	BT2																					
	c	How will you solve an L.P.P graphically? Give example.	05	CO1	BT2																					
Q.2	a	Find the value of the game. <table><tr><td colspan="2" rowspan="2"></td><th colspan="3">Player B</th></tr><tr><th>I</th><th>II</th><th>III</th></tr><tr><th rowspan="3">Player A</th><th>I</th><td>-3</td><td>-2</td><td>6</td></tr><tr><th>II</th><td>2</td><td>0</td><td>2</td></tr><tr><th>III</th><td>5</td><td>-2</td><td>-4</td></tr></table>			Player B			I	II	III	Player A	I	-3	-2	6	II	2	0	2	III	5	-2	-4	05	CO4	BT3
		Player B																								
		I	II	III																						
Player A	I	-3	-2	6																						
	II	2	0	2																						
	III	5	-2	-4																						
	b	In a bank cheques are cashed at a single teller counter customers arrive at counter in poisson's manner at average rate of 30 customers per hour. The teller takes average of 1.5 minute to make cheque: a) Calculate percentage of time teller is busy. b) Calculate average time customers is expected to wait.	05	CO4	BT2																					
Q.3		Solve the following problem using the Graphical method, Maximize $Z = 4x_1 + 6x_2$, Subject to, $x_1 + x_2 \geq 8$ $6x_1 + x_2 \geq 12$, Where, $x_1, x_2 \geq 0$.	10	CO1	BT2																					
Q.4		Solve the following problem using the Simplex method, Maximize $Z = 3x_1 + 5x_2 + 4x_3$ Subject to, $2x_1 + 3x_2 \leq 8$, $2x_1 + 5x_3 \leq 10$, $3x_1 + 2x_2 + 4x_3 \leq 15$, Where, $x_1, x_2, x_3 \geq 0$.	10	CO1	BT3																					

Q.5	Machines'	Solve the problem of assigning five operators of five machines. The assignment cost is given below: <table><tr><td></td><th colspan="5">Operators</th></tr><tr><td></td><th>I</th><th>II</th><th>III</th><th>IV</th><th>V</th></tr><tr><th>A</th><td>10</td><td>5</td><td>13</td><td>15</td><td>16</td></tr><tr><th>B</th><td>3</td><td>9</td><td>18</td><td>3</td><td>6</td></tr><tr><th>C</th><td>10</td><td>7</td><td>2</td><td>2</td><td>2</td></tr><tr><th>D</th><td>5</td><td>11</td><td>9</td><td>7</td><td>12</td></tr><tr><th>E</th><td>7</td><td>9</td><td>10</td><td>4</td><td>12</td></tr></table>		Operators						I	II	III	IV	V	A	10	5	13	15	16	B	3	9	18	3	6	C	10	7	2	2	2	D	5	11	9	7	12	E	7	9	10	4	12	10	CO4	BT3
	Operators																																														
	I	II	III	IV	V																																										
A	10	5	13	15	16																																										
B	3	9	18	3	6																																										
C	10	7	2	2	2																																										
D	5	11	9	7	12																																										
E	7	9	10	4	12																																										
Q.6		Solve Maximize the following assignment problem <table><tr><td></td><th>I</th><th>II</th><th>III</th><th>IV</th><th>V</th></tr><tr><th>A</th><td>62</td><td>78</td><td>50</td><td>101</td><td>8</td></tr><tr><th>B</th><td>72</td><td>84</td><td>61</td><td>73</td><td>5</td></tr><tr><th>C</th><td>87</td><td>92</td><td>111</td><td>71</td><td>8</td></tr><tr><th>D</th><td>48</td><td>64</td><td>87</td><td>77</td><td>8</td></tr><tr><td></td><td></td><td></td><td></td><td></td><td>0</td></tr></table>		I	II	III	IV	V	A	62	78	50	101	8	B	72	84	61	73	5	C	87	92	111	71	8	D	48	64	87	77	8						0	10	CO4	BT3						
	I	II	III	IV	V																																										
A	62	78	50	101	8																																										
B	72	84	61	73	5																																										
C	87	92	111	71	8																																										
D	48	64	87	77	8																																										
					0																																										
Q.7		A manufacturer produces two types of models M1 and M2. Each model of the type M1 requires 4 hours of grinding and 2 hours of polishing; whereas each model of the type 2 hour of grinding and 5 hours of polishing. The manufacturer has 2 grinder and 3 polishers. Each grinder works 40 hours a week and each polisher work for 60 hours a week. Profit on M1 model is Rs 3 and on model M2 is Rs 4 whatever is produced in a week is sold in the market. How should the manufacturer allocate his production capacity to the two types of models so that he makes the maximum profit in a week?	10	CO1	BT2																																										

== ALL THE BEST ==

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyse, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Conceptualize and formulate various real-life problems.
CO 2	Learn various analytical techniques to evaluate the problems.
CO 3	Use software tool for solving LPP.
CO 4	Generate optimal solutions for various queuing, transportation, assignment and dynamic programming problems



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Production Engineering
In-Semester Examination / Evaluation II
Academic Year: 2025-26 (Semester I)

Class: TY B.Tech. Course: Metal Cutting Theory Course Code: PCC-PE305
Date: 17.11.2025 Time: 10:00 to 12:00 Max. Marks: 60

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.	Questions	Marks	CO	BT
Q.1	A With the help of diagram discuss in details chip generation in machining, and chip breaker.	6	CO1	BT2
Q.2	A In an orthogonal cutting process, the following observations were made: Depth of cut = 0.25 mm, Chip thickness ratio = 0.45, Width of cut = 4 mm, Cutting velocity = 40 m/min Cutting force component parallel to cutting velocity vector = 1150 N Cutting force component normal to cutting velocity vector = 140 N Rake angle = 18° Determine the resultant cutting force, power of cutting, shear plane angle, friction angle and force component parallel to shear plane.	8	CO5	BT5
	B What is Taylor's tool life equation? With the help suitable graphs explain tool life variations.	6	CO2	BT2
Q.3	Suppose a turning operation is to be performed with HSS tooling on mild steel, with Taylor tool life parameters $n=0.125$, $C=70$ m/min. Workpart length = 500mm and diameter = 100 mm. Feed = 0.25 mm/rev. Handling time per piece = 5.0 min, and tool change time = 2.0 min. Cost of machine and operator = \$30/hr, and tooling cost = \$3 per cutting edge. Find: (a) cutting speed for maximum production rate, and (b) cutting speed for minimum cost (c) the hourly production rate and cost per piece for the two cutting speeds computed.	10	CO1	BT4
Q.4	Describe any two modern cutting tool materials used in metal cutting. For each material, write their important properties and their particular machining application.	8	CO2	BT3
	A What are considerations in designing Form Tool in bending? Write along with formulas.	6	CO3	BT2
Q.5	A cup drawing operation is performed using a steel sheet having the following properties and dimensions: - Desired cup inside diameter, = 100 mm - Desired cup height, = 60 mm - Sheet thickness, = 2.5 mm - Punch and die corner radii, = 5 mm - Ultimate tensile strength, = 420 MPa - Yield strength, = 200 MPa - Blank diameter, = 180 mm	6	CO3	BT5

	Assume that the material flow and friction conditions are similar to standard deep drawing of mild steel. Determine: (a) The drawing ratio (DR) (b) The percentage reduction in diameter (R) (c) The maximum drawing force (F)			
Q.6	Design a suitable single-point cutting tool for this turning operation. A mild steel workpiece of $\text{Ø}60 \text{ mm} \times 150 \text{ mm}$ is turned at 500 rpm with feed = 0.25 mm/rev and depth of cut = 2 mm. Using this data: a. Specify the tool signature values for each angle and nose radius: Give justification for each value based on mild-steel machining. b. Draw a neat sketch of the tool showing and labelling: c. Select the most suitable tool material with justification: d. Recommend a tool holder / clamping arrangement e. Calculate the approximate cutting force using suitable assumptions and comment briefly on tool rigidity requirements.	10	CO4	BT6

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	The students shall have the knowledge of fundamentals of metal cutting technology including the process, measurements, design and selection of various cutting tools and their industrial specifications.
CO 2	Select suitable cutting tool for performing identified machining operation.
CO 3	Design form tool for given application and conditions.
CO 4	Design single and multiple point cutting tools for given conditions and applications.
CO 5	Understand cutting forces generated during different cutting operations.



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded
Department of Production Engineering
Academic Year: 2025-26 (semester I)
End Term Examination

Class: Third Year

**Course Name: Production
Planning & Control**

Course Code: PCC-PE303

Date: 26.11.2025

Time: 10:00 to 12:00

Max. Marks: 60

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iv. Figures to right indicate full marks.
- v. The symbols carry usual meaning.

Q.1		Solve the following questions. (Any FOUR)	Marks	CO	BT
	a	Enlist the types of flow system and explain all types.	05	CO3	BT1
Q.1	b	Explain with neat sketch process layout.	05	CO2	BT2
	c	The following data are given for a company estimated output = 80,000 units, Fixed cost = Rs. 4,00,000 Variable cost = Rs. 10 per unit Selling price = Rs. 20 per unit. Find out the BEP analytically and graphically.	05	CO2	BT3
	d	What is the role of marketing aspects in product analysis.	05	CO2	BT1
	e	Explain Types of production systems.	05	CO1	BT2
Q.2		Solve the following questions. (Any FOUR)			
	a	Explain why an organization has to go for forecast? Explain in the brief any two forecasting methods.	05	CO2	BT2
	b	Define Inventory. Discuss various types of inventory costs.	05	CO4	BT1
	c	Explain the significance and construction of the process chart.	05	CO3	BT2
	d	What do you mean by aggregate planning? What are the guidelines for aggregate planning?	05	CO5	BT2
	e	Define facility layout. What are the objectives of good plant layout?	05	CO3	BT1
Q.3		Solve the following questions. (Any TWO)			
	a	A particular item has a demand of 9,000 units/year. The cost of one procurement is Rs. 100 and the holding cost per unit is Rs. 2.40 per year. The replacement is instantaneous, and no shortages are allowed. Determine (i) the economic lot size, (ii) the number of orders per year, (iii) the time between orders, (iv) the total cost per year if the cost of one unit is Re. 1.	10	CO4	BT3
	b	Draw the two-handed process chart for assembly of nut and bolt.	10	CO3	BT6

c	There are seven jobs of which has to go through the machines A & B in the order of AB. Processing times in hours are given as							10	CO5	BT5	
	Job	1	2	3	4	5	6				7
	Machine A	4	13	16	7	11	12				10
	Machine B	9	11	11	7	13	2				4
	Determine a sequence of these jobs that will minimize the total elapsed time.										

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Discuss the different types of production systems.
CO 2	Analyze the product development and design technique.
CO 3	Differentiate between production layouts.
CO 4	Understand the production order procedure.
CO 5	Construct and model Aggregate production plans.

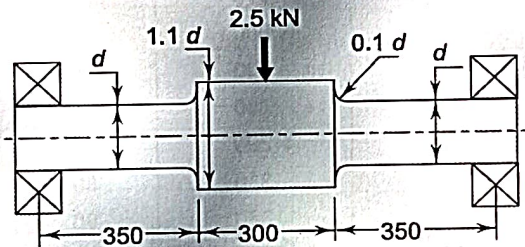


Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Production Engineering
End Semester Examination
Academic Year: 2025-26 (semester I)

Class: TY B.Tech.	Course: Machine Design	Course Code: PCC-PE302
Date: 19.11.2025	Time: 10:00 to 12:00	Max. Marks: 60

Note:

- Attempt all the questions.
- Refer the standard table for the shaft diameter printed on back side of the question paper.
- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.
- Use of non-programmable calculator is allowed.

Q.N.		Marks	CO	BT
	Solve the following questions.			
Q.1	A non-rotating shaft supporting a load of 2.5 kN is shown in Fig. The shaft is made of brittle material, with an ultimate tensile strength of 300 N/mm². The factor of safety is 3. Determine the diameter of the shaft. [Given, $K_t = 1.61$] 	10	CO4	BT3 ,4& 5
Q.2	Briefly explain about Key and types of Keys. Design the rectangular key for a shaft of 55 mm diameter. The shearing and crushing stresses for the key material are 48 MPa and 80 MPa.	20	CO3	BT2 ,5
Q.3	Explain Muff Coupling with labelled diagram. Write the advantages and disadvantages of the Muff Coupling. Design a muff coupling which is used to connect two steel shafts transmitting 40 kW at 350 r.p.m. The material for the shafts and key is plain carbon steel for which allowable shear and crushing stresses may be taken as 40 MPa and 80 MPa respectively. The material for the muff is cast iron for which the allowable shear stress may be assumed as 15 MPa.	30	CO5	BT1 ,5& 6

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO1	Understand basic design procedure and the necessary design considerations.
CO2	Define and distinguish static and fluctuating loads/stresses.
CO3	Derive empirical relations for different machine elements using relevant theories of failure.
CO4	Analyse behaviour of machine components under static and fluctuating loads/stresses.
CO5	Design shafts, gears, clutches, power screws etc. for real life applications when subjected to static loading using generative design.

Refer the standard table for the shaft diameter.

Table 13.1. Proportions of standard parallel, tapered and gib head keys.

Shaft diameter (mm) upto and including	Key cross-section		Shaft diameter (mm) upto and including	Key cross-section	
	Width (mm)	Thickness (mm)		Width (mm)	Thickness (mm)
6	2	2	85	25	14
8	3	3	95	28	16
10	4	4	110	32	18
12	5	5	130	36	20
17	6	6	150	40	22
22	8	7	170	45	25
30	10	8	200	50	28
38	12	8	230	56	32
44	14	9	260	63	32
50	16	10	290	70	36
58	18	11	330	80	40
65	20	12	380	90	45
75	22	14	440	100	50



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Production Engineering.
End-Semester Examination / Evaluation
Academic Year: 2025-26 (semester I)

Class: TY B.Tech.

Course: Renewable Energy
Resources.

Course Code: OE-PE301

Date: 01.12.2025

Time: 10:00 to 12:00

Max. Marks: 60

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.		Questions	Marks	CO	BT
Q.1	a	Define the terms: i. Energy intensity ii. Energy-GDP elasticity iii. Cogeneration.	06	CO1	BT1
	b	Find the day-length in hours at New Delhi ($28^{\circ}35' N$, $77^{\circ}12' E$) on July 1 for a south-facing surface tilted at 10° .	08	CO2	BT3
Q.2	a	Explain in detail the different modes of wind power generation.	06	CO3	BT2
	b	State and Explain the different types of tidal power plants with neat sketches.	08	CO4	BT2, BT3
Q.3	a	Discuss the utilisation of geothermal energy in India for different purposes.	08	CO4	BT2
	b	Explain the different types of wave energy conversion machines. Describe their construction and working principles with sketch.	08	CO4	BT2, BT3
Q.4	a	Explain the construction and working principle of a hydrogen-oxygen acidic fuel cell.	06	CO5	BT2
	b	Describe and sketch the different types of fuel cells with their electrolytes, operating temperatures, and applications.	10	CO5	BT2, BT3

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Interpret the energy reserves of India and the potential of different energy sources.
CO 2	Measure the solar radiation parameters and performance of different solar collectors.
CO 3	Calculate different parameters of the wind turbine rotor.
CO 4	Implicit the importance and applications of geothermal and ocean energy.
CO 5	Demonstrate knowledge in the field of fuel cells and potential for power generation.



Shri Guru Gobind Singh Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Production Engineering
Academic Year: 2025-26 (semester I)
End Semester Examination

Class: TY B.Tech.

Course Name: Industrial Engg.

Course Code: PEC-PE301

Date: 21.11.2025

Time: 10:00 to 12:00

Max. Marks: 60

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iv. Figures to right indicate full marks.
- v. The symbols carry usual meaning.

Q.N.		Marks	CO	BT																									
Q.1	Attempt any three questions from the following sub questions.		12																										
	a	Explain the principles of labour legislation.	4	CO5	BT1																								
	b	Discuss forecasting by exponential smoothing and find forecast sales for the next period in the following example: Forecast sales for last period = 544, Actual sales for last period = 542, Assume $\alpha = 0.2$.	4	CO4	BT5																								
	c	What do you understand by ergonomics. What is role of industrial engineer to productivity improvement?	4	CO1	BT2																								
	d	Differentiate between production and productivity.	4	CO1	BT1																								
Q.2	Answer any three questions from the following.		18																										
	a	Discuss in detail the multiple activity chart.	6	CO2	BT4																								
	b	Calculate the number of machines required to manufacture three components from the following data: <table border="1"><thead><tr><th>Particulars</th><th colspan="3">Components</th></tr><tr><td></td><th>A</th><th>B</th><th>C</th></tr></thead><tbody><tr><td>(1) Setting time per lot</td><td>27 min</td><td>57 min</td><td>40 min</td></tr><tr><td>(2) Operating cycle time/piece</td><td>1.8 min</td><td>3 min</td><td>2 min</td></tr><tr><td>(3) Lot size</td><td>350</td><td>500</td><td>550</td></tr><tr><td>(4) Production per month</td><td>2000</td><td>4500</td><td>3000</td></tr></tbody></table> The plant is working for 45 hrs per week.	Particulars	Components				A	B	C	(1) Setting time per lot	27 min	57 min	40 min	(2) Operating cycle time/piece	1.8 min	3 min	2 min	(3) Lot size	350	500	550	(4) Production per month	2000	4500	3000	6	CO4	BT5
Particulars	Components																												
	A	B	C																										
(1) Setting time per lot	27 min	57 min	40 min																										
(2) Operating cycle time/piece	1.8 min	3 min	2 min																										
(3) Lot size	350	500	550																										
(4) Production per month	2000	4500	3000																										
	c	Give the advantages and disadvantages of merit rating.	6	CO5	BT2																								
	d	Explain the two basic Gantt charts for progress control.	6	CO4	BT3																								

Q.3	Solve any three numerical or expressions from the following.		30		
	a	Determine the economical machine using the following data:			
		Machine particulars	Centre lathe	Auto lathe	
		(1) Set up time	0.5 hrs	6 hrs	
		(2) Operating time	35 min	2 min	
		(3) Set up labour cost	₹ 7 per hr	₹ 7 per hr	
		(4) Operating labour cost	₹ 6 per hr	₹ 2 per hr	
		(5) Tooling cost	Nil	₹ 200	
		(6) Machine over-heads	₹ 10 per hr	₹ 28 per hr	
		(7) Lot size	1500 Nos.	1500 Nos.	
	b	Describe the sequence of motion for signing a letter with the definition of motion and symbols using Therblig elements.			
	c	Find the expression for finding E.O.Q. by mathematical means.			
	d	Explain the value analysis procedure with the advantages and applications of it.			

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 – Remember / List, BT2 – Understand / Summarize, BT3 – Apply / Construct / Organize, BT4 – Analyse / Function / Compare, BT5 – Evaluate / Estimate / Recommend, BT6 - Create / Build / Formulate.

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Evaluate the productivity and implement various productivity improvement techniques.
CO 2	Apply work study techniques and understand its importance for better productivity.
CO 3	Demonstrate the ability to select plant location, appropriate layout and material handling equipment.
CO 4	Use of Production planning and control tools for effective planning, scheduling and managing the shop floor control.
CO 5	Plan inventory requirements and exercise effective control on manufacturing requirements.



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Production Engg.
End Semester Examination
Academic Year: 2025-26 (semester I)

Class: TY B.Tech.

Course: Mechanical Working of Metals

Course Code: PCC-PE301

Date: 19.11.2025

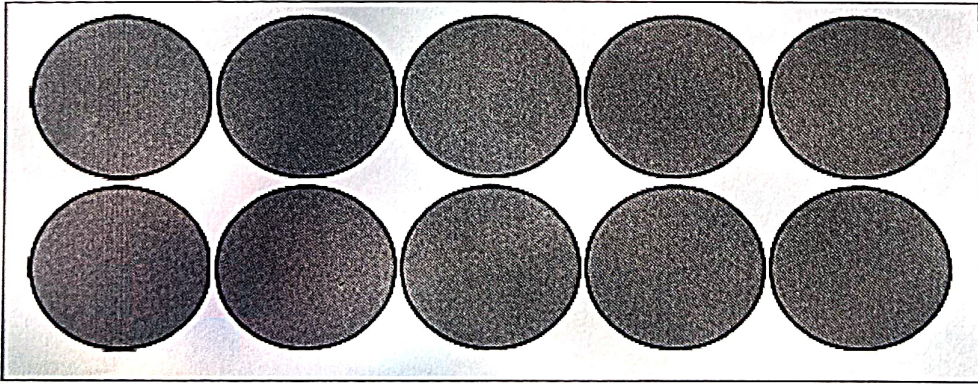
Time: 10:00 to 12:00

Max. Marks: 60

Note:

Reg.No.....

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Solve **any SIX** questions from questions 1 to 7.
- iii. Figures to right indicate full marks.
- iv. Paper containing **2 page..**

Q.N.			Marks	CO	BT
Q. 1		Explain the forging process with a neat sketch. What are the different types of forging methods?	10	CO2	BT2
Q. 2		Describe the extrusion process, its types, and key advantages.	10	CO1	BT3
Q. 3		What is wire drawing process? Also, explain the different zones present in the wire drawing with neat sketch.	10	CO2	BT2
Q. 4		Explain Compound die and Progressive die with neat sketch.	10	CO2	BT5
Q. 5		Compare hot working and cold working processes based on surface finish, dimensional accuracy, energy requirement, and mechanical properties.	10	CO4	BT4
Q. 6	A	A solid cylinder stainless steel work piece of 200mm diameter and 150 mm height with flat die in open die forging is reduced by 50 % in height. Assuming the flow stresses of material as 1000 Mpa and the COF to be 0.2. Calculate the estimated forging force at the end of the stroke.	05	CO3	BT2
	B	Using direct extrusion process, a round billet of 15 cm length and 7.5 cm diameter is extruded. Consider an ideal deformation process (no friction and no redundant work), extrusion ratio 6 and average flow stress of material 120 MPa. Calculate the pressure (MPa) on the ram.	05	CO3	BT3
Q. 7	A	Calculate the total number of circular blanks produced from a long 1000 mm sheet of diameter 140 mm with a margin of 2 mm. Also, find out the percentage utilization of the sheet. 	05	CO3	BT5
	B	Calculate the number of draws required for the cup of diameter 35 mm and height	05	CO3	BT3

		of 210 mm. If the draw ratio for the first stage is 1.6 and for second stage is 1.5, for third and subsequent it is 1.3.			
--	--	--	--	--	--

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Classify different metal forming processes.
CO 2	Design dies for cutting, forging, bending, drawing, piercing, and blanking
CO 3	Compute material properties, forces and defects in metal forming processes.
CO 4	Select equipment for metal forming processes.